

IUBAT — INTERNATIONAL UNIVERSITY OF BUSINESS

AGRICULTURE AND TECHNOLOGY

Founded 1991 by Md. Alimullah Miyan

IUBAT School of Computer Science and Engineering
Department of Computer Science and Engineering

Mid Term Examination of Summer Semester 2026

Program: Bachelor of Computer Science and Engineering (BCSE)

Course Instructor: Ferdaus Anam Jibon

Course Code: CSE 4385	Time: 01 Hour 30 Minutes
Course Title: Computer Vision and Image Processing	Full Marks: 100
Section: E & F	Date: 05 – 07 – 2026

Instructions:

1. Answer all the questions. 2. Marks allocated are indicated in the right margin.

CO	1	2	3
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PO	a	b
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1. Consider the 8-bit grayscale image shown in Figure 1, represented by the matrix given beside. Apply the histogram equalization shifting technique to this image and determine the resulting pixel value of the 3rd row after the transformation. **[CO2, C3, PO1 — 25 marks]**



50	56	60	62	100	68
52	72	70	120	140	90
54	78	136	96	150	76
110	120	132	128	96	90
98	140	148	124	130	70
99	154	86	88	80	145

Figure 1: Football Ground and Corresponding Image Matrix

2. Consider the following 3×3 matrix, which represents the RGB values corresponding to Figure 1. Convert the RGB value of the first pixel of the matrix into its equivalent HSI (Hue, Saturation, Intensity) and CMY (Cyan, Magenta, Yellow) representations. Justify your answer with an appropriate mathematical derivation. **[CO2, C3, PO1 — 15 + 10 = 25 marks]**

(120, 150, 180)	(130, 0, 0)	(0, 100, 0)
(200, 0, 0)	(0, 150, 90)	(0, 0, 180)
(255, 45, 0)	(100, 0, 180)	(0, 90, 255)

3. A university office is preparing to scan and store three documents: a black-and-white attendance sheet, a colored event photo, and a student's ID card scan. The university authority has decided to prepare a report for technical specifications and future reusability, the report should i) explain the different types of digital images that may correspond to the files described above, ii) explain the technique used for representing digital images in computers, and iii) state the commonly used digital image formats suitable for storing and processing each of the above files. Write and prepare the report for the university. **[CO1, C2, PO1 — 8 + 10 + 7 = 25 marks]**

4. Digital Image Processing consists of five fundamental steps: Image Formation, Image Acquisition, Image Transformation, Feature Selection, and Classification. Briefly explain the core principles of Image Transformation and Feature Selection, and analyze the commonly used techniques and their applications in enhancing image analysis, pattern recognition, and classification tasks associated with these two steps. **[CO1, C2, PO1 — 10 + 15 = 25 marks]**

Bloom's Taxonomy Levels in Cognitive Domain:

C1: Remember; C2: Understand; C3: Apply; C4: Analyze; C5: Evaluate; C6: Create

Course Outcomes:

CO1: Apply digital image processing techniques and computer vision algorithms to solve image analysis, identification, recognition, segmentation, detection, and classification operations.

CO2: Explain the fundamental concepts of computer vision and digital image processing for optimized solution, interpretation, and decision making.

Program Outcomes:

PO1: Apply knowledge of mathematics, natural science, engineering fundamentals, and an engineering specialization as specified in K1 to K4 respectively to the solution of complex engineering problems.